

GenAI + ML + Deep Learning Project Roadmap

Strategy

This roadmap helps you progress from foundational skills to advanced AI systems.

- 1 Foundation: Python, ML basics, data handling
- 2 Modern AI: LLMs, embeddings, RAG systems
- 3 Advanced: Deep learning, TensorFlow, production systems

Phase 1 — LLM + GenAI

Start by building practical GenAI applications.

- 1 AI Resume & Job Matcher (LLM + Streamlit)
- 2 PDF Chatbot using RAG (embeddings + FAISS)
- 3 Smart SQL Assistant using LLM

Phase 2 — Machine Learning & Deep Learning

Focus on real ML understanding and pipelines.

- 1 End-to-End ML Pipeline (data → model → evaluation)
- 2 Neural Network from scratch (NumPy)
- 3 Deep Learning project (TensorFlow CNN or NLP)

Phase 3 — Advanced AI Systems

Build industry-level systems combining AI + ML.

- 1 Production RAG system (vector DB + retrieval)
- 2 Hybrid ML + LLM system (e.g., fraud detection + explanation)
- 3 Multi-agent AI system
- 4 Real-time data + AI system

What Makes a Project Advanced

- 1 Clear problem statement
- 2 Data pipeline and preprocessing
- 3 Model training and evaluation
- 4 Deployment or UI (Streamlit/Web)
- 5 Business or real-world impact explanation

Timeline

- 1 Phase 1: 2–3 weeks
- 2 Phase 2: 3–4 weeks
- 3 Phase 3: 4–6 weeks